American Computer Science League

2021-2022 ● Contest 1: Short Problem Solutions ● Classroom Division

1. Computer Number Systems

$2122_{10} = 4 * 8^3 + 1 * 8^2 + 1 * 8^1 + 2 * 8^0$

$= 4112_8$

D. 4112

2. Computer Number Systems

Fibonacci sequence 0, 1, 1, 2, 3, 5, 8, 13, … in base 10
Fibonacci sequence 0, 1, 1, 10, 11, 101, 1000, 1101, ... in binary

1st: 10 from 2
2nd: 101 from 5
3rd: 1000 from 8
4th: 1101 from 13

B. 13

3. Recursive Functions

$f(10) = f(10-2) + 3 = f(8) + 3 = 9 + 3 = 12$
$f(8) = f(8-2) + 3 = f(6) + 3 = 6 + 3 = 9$
$f(6) = f(6-1) - 1 = f(5) - 1 = 7 - 1 = 6$
$f(5) = f(5-1) - 1 = f(4) - 1 = 8 - 1 = 7$
$f(4) = f(4-1) - 1 = f(3) - 1 = 9 - 1 = 8$
$f(3) = 3^2 = 9$

E. 12

4. Recursive Functions

$f(20,12) = f(20-3,12+1) + 3 = f(17,13) + 3 = 60 + 3 = 63$
$f(17,13) = f(17-3,13+1) + 3 = f(14,14) + 3 = 57 + 3 = 60$
$f(14,14) = 2 * f(14+2,14-3) - 1 = 2 * f(16,11) - 1$
$= 2 * 29 - 1 = 57$
$f(16,11) = f(16-3,11+1) + 3 = f(13,12) + 3 = 26 + 3 = 29$
$f(13,12) = f(13-3,12+1) + 3 = f(10,13) + 3 = 23 + 3 = 26$
$f(10,13) = 10 + 13 = 23$

C. 63

5. What Does This Program Do? (Branching)

The following table can be used to trace the program:

a	b	c	d
0	2	4	6
0	2	12	6
0	4	12	6
0	4	12	6
0	10	12	6

c / d − b + a ^ 3 * d
= 12 / 6 −10 + 0 ^ 3 * 6
= 2 − 10 + 0 * 6
= 2 − 10 +0
= -8

A. -8

6. Computer Number Systems

$2122_{10} = 8 * 16^2 + 4 * 16^1 + 10 * 16^0$
$= 84A_{16}$

D. 84A

7. Computer Number Systems

Fibonacci sequence 0, 1, 1, 2, 3, 5, 8, 13, … in base 10
Fibonacci sequence 0, 1, 1, 2, 3, 5, 10, 15, ... in octal

1st: 1 from 1
2nd: 1 from 1
3rd: 10 from 8
4th: 15 from 13
5th & 6th: 131 from 89

C. 89

8. Recursive Functions

$f(20) = 2 * f(20 - 3) + 1 = 2 * f(17) + 1 = 2 * 19 + 1 = 38 + 1 =$

$f(17) = 2 * f(17 - 3) + 1 = 2 * f(14) + 1 = 2 * 9 + 1 = 18 + 1 =$
$f(14) = [f(14 + 2)/3] = [f(16)/3] = [29/3] = 9$

$f(16) = 2 * f(16 - 3) + 1 = 2 * f(13) + 1 = 2 * 14 + 1 = 28 + 1 =$
$f(13) = [f(13 + 2)/3] = [f(15)/3] = [43/3] = 14$

$f(15) - 2 * f(15 - 3) + 1 = 2 * f(12) + 1 = 2 * 21 + 1 = 42 + 1 =$
$f(12) = 2 * 12 - 3 = 24 - 3 = 21$

B. 39

9. Recursive Functions

$f(20) = f(18/2) - 1 = f(9) - 1 = 18 - 1 = 17$
$f(9) = 2 * 9 = 18$
$f(17) = 2 * 17 = 34$
$f(34) = f(32/2) - 1 = f(16) - 1 = 13 - 1 = 12$
$f(12) = f(10/2) - 1 = f(5) - 1 = 10 - 1 = 9$

Therefore: $f(f(f(f(20)))) = f(f(f(17)))$
$= f(f(34))$
$= f(12)$
$= 9$

E. 9

10. What DoesThis Program Do? (Branching)

The following table can be used to trace the program:

a	b	c	d
10	4	0	6
10	4	0	4
10	4	1	4
10	4	2	4
1	4	2	4
1	4	2	2

x = a − b * c + d
= 1 − 4 * 2 + 2
= 1 − 8 + 2
= -7 + 2
= -5

D. -5